



FALL 2026: Tuesdays and Thursdays from 15:00-16:15

POPH 001-011 (CRN 44210, 4 credits) — Lewis Lab 316

August 24, 2026 until Friday, December 4, 2026. Schedule subject to change
(last modif.: August 23, 2026)



Instructor: Eric Mathieu Delmelle — Email: erd424@lehigh.edu

Student drop-in hours: Tuesdays 10.30 am

Virtual (zoom) appointments: email me please.

Office: HST - 139.

1 Catalog Description

Despite significant advances in medicine and public health, inequities in health persist. Understanding health on a population level is an approach that seeks to improve the health of the whole population, unravel variations in health outcomes, and to identify effective strategies for reducing or eliminating inequities. The purpose of this course is to provide students with an understanding of: 1) how population and public health are defined and measured; and 2) the determinants of population health.

2 Course Learning Outcomes

Upon completion of this course, students will be able to:

1) Understand Core Concepts in Population Health:

- Define key concepts such as population health, health equity, and social determinants of health.

2) Apply Measurement Techniques:

- Utilize foundational epidemiological measures (e.g., incidence, odds ratio, prevalence) and demographic tools to assess health outcomes and population dynamics.

3) Analyze Determinants of Population Health:

- Explore individual, interpersonal, community, and macrosystem-level determinants of health.
- Evaluate multilevel interactions between genetic, environmental, and behavioral factors over the life course.

4) Develop Evidence-Based Interventions:

- Propose and evaluate strategies for improving health outcomes using prevention, health promotion, and risk reduction principles.

5) Demonstrate Research and Communication Skills:

- Design and conduct studies employing sound research methodologies.
- Present findings through effective written, oral, and visual communication.

6) Address Health Inequities in Vulnerable Populations:

- Identify health disparities among diverse populations and propose culturally sensitive interventions.

3 Course Materials

The materials for our course come directly from the textbook by Young, T. K. (2004). **Population health: concepts and methods**, *Oxford University Press*. The book is available in the bookstore, but you can also find electronic copies of each chapter, uploaded on <https://coursesite.lehigh.edu> (note that the electronic copies are not as "nice" as the book). In addition to your textbook, you may be asked to read relevant articles (e.g. press, journal), which will be available online.

4 Grading Scale

Table 1 summarizes the correspondence between the percentage of grades, total points, and letter grades. You can earn up to 300 points in this course. There will be opportunities in the quizzes and in the assignments to get extra points.

Table 1: Grading protocol

Percentage	Points	Letter Grade
$\geq 93\%$	279+ pts	A
$\geq 90\%$	270+ pts	A-
$\geq 87\%$	261+ pts	B+
$\geq 83\%$	249+ pts	B
$\geq 80\%$	240+ pts	B-
$\geq 77\%$	231+ pts	C+
$\geq 73\%$	219+ pts	C
$\geq 70\%$	210+ pts	C-
$\geq 67\%$	201+ pts	D+
$\geq 63\%$	189+ pts	D
$\geq 60\%$	180+ pts	D-
$< 60\%$	< 180 pts	F

Table 2: Type of assignment, percentage, and due dates

Assessment Methods (300 points total)	Activity	Grade Percent (100% total)	Due Date
Assignments (60 points) (12 pts each)	Solve the Questions	20%	Assignment 1: 9/11 (Fri) Assignment 2: 9/29 (Tue) Assignment 3: 10/7 (Wed) Assignment 4: 10/21 (Wed) Assignment 5: 11/10 (Tue)
Quizzes (35 pts; 5 pts each) (7 quizzes; none dropped — all count)	In-class Quiz (MC/TF) 5–7.5 min 1-page cheat sheet allowed	≈12%	Quiz 1: 9/8 (Tue) Quiz 2: 9/15 (Tue) Quiz 3: 10/1 (Thu) Quiz 4: 10/15 (Thu) Quiz 5: 10/20 (Tue) Quiz 6: 11/10 (Tue) Quiz 7: 11/19 (Thu)
Midterm Exam (50 points)	In-class exam full 75 min no notes	≈17%	Tuesday 10/27 (entire class period) Covers Chapters 1–5
Health News Presentation (5 points)	Individual presentation ≈2 min each	≈1.7%	Once per semester
Individual Paper (40 points)	Community Assessment	≈13%	Final Paper: 11/24
Peer Presentation (15 points)	Present peer community	5%	Slides due: 10/5 (midnight) Presented 10/6 & 10/8
Group Work (65 points)	Presentation (15 min) & Report (8–12 pages)	≈22%	Topics assigned: 9/1 Scoping note: 9/30 Checkpoint report: 11/12 Slides due: 11/30 (midnight) Presentations: 12/1 & 12/3 Report due: 12/7
Final Exam (30 points)	Online exam cumulative	10%	COURSESITE December 8–16, 2026

5 Evaluation and Grading

Everything in this course is due at midnight on the date listed — that is, 11:59 pm EDT/EST at the end of that day. This holds without exception: homework, papers, group reports, and presentation slides alike. There are no noon deadlines, no morning deadlines, and nothing that is due at a different hour because it happens to be a different kind of work. If a date appears next to a deliverable anywhere in this syllabus, you have until the end of that day. Earlier is always fine.

All seven quizzes count — none is dropped. Table 2 summarizes the weights of the various assignments and due dates.

6 Late Assignments/Papers:

Assignments are due before or on their due dates. Grades for assignments will be **reduced by 20% per day**, unless the due date has been discussed with the instructor ahead of the due date (Each student should contact the instructor in advance to arrange when an assignment is going to be late -e.g. conferences, travel plans, etc..). **Quizzes, the midterm exam, and the final exam are NOT available after their scheduled time**, except for students with a documented excuse of the kind described above, or who require extended time or a different setting. Medical emergencies may arise, and in these situations, I will not deduct points for the lateness if you provide a physician's note and if the assignment is submitted within 7 days (one week) of the deadline (weekends are considered as working days in this respect).

7 Assignments (60pts, 20%)

The assignments are aimed to assess your comprehension of basic population health concepts and issues. There will be five assignments (homeworks) in total, **12 points each**. **All assignments are required to be completed**. Assignments are submitted to the Course Site. I will also assess your writing skills (spelling, grammar, and syntax). In case of essay questions, your response should be in a **double-spaced paper using 12-pt. Times Roman or Arial font with 1-inch page margins**. All assignments should be in Word format (i.e., doc or docx) or Google Docs, or Overleaf documents.

We try to follow this protocol for naming assignments in this course: Your assignment file includes your last name, first initial, and the number given to the assignment; Example filename: **[DELMELLE_EM_Assignment1.docx]**. Please write your name on your assignment.

Assignment Sequence:

- **Assignment 1 (Due 9/11):** Social & Environmental Determinants – establishes the structural framework early.
- **Assignment 2 (Due 9/29):** Population Health Theories / Socio-Ecological Model – provides organizing framework.

- **Assignment 3 (Due 10/7):** Comparing Health Events in Populations – measurement becomes interpretable.
- **Assignment 4 (Due 10/21):** Epidemic Investigation & Test Characteristics – measurement tools in context.
- **Assignment 5 (Due 11/10):** Assessing Causation (Bradford Hill, confounding, bias).

Access **Course Site** by going to <https://coursesite.lehigh.edu> and log in with your Lehigh credentials. For technical assistance, contact the Library and Technology Services help desk.

8 Health News Presentation (5pts, $\approx 1.7\%$)

Each student will present **once** during the semester at the beginning of class. **Three students present per session**, each bringing something different.

What we are looking for. Find something happening in the world — or right here, locally — that **tells us something about a population**. Not simply a health story, but one that reveals how a population lives, how it is changing, or why one group fares differently from another. A new outbreak, a shift in life expectancy, a policy announcement, a set of census or surveillance figures, a pattern you have noticed in your own town: all of these qualify, provided you can say what they show us about the people behind the numbers.

The point is to train the reflex of reading the news the way a population health researcher does — asking *who* is affected, *why them*, and *what would have to change*. Over the semester, these two-minute contributions build a shared and current picture of the field alongside the textbook.

Who presents when. The presenter schedule is drawn up once enrolment is settled and posted on Course Site in **Week 2**; the course schedule at the end of this syllabus marks which sessions carry a Health News slot, but not yet whose name goes in it. You will know your date at least three weeks in advance, and you may swap with a classmate provided you both tell me beforehand.

Format:

- Duration: **about 2 minutes**
- **Three presenters per session** (≈ 6 minutes total).
- Present a current health event, news item, or demographic pattern that interests you — and say, in one sentence, what it tells us about the population concerned.

Topics may include:

- Recent disease outbreaks or epidemiological developments.
- New political announcements related to health policies.
- Recently released health statistics or demographic data
- New public health measures or interventions.
- Emerging health trends in specific countries or populations.

- Notable or concerning demographic patterns.
- Any other health-related current event or topic of interest.

These presentations will help you stay anchored to real-world applications of course concepts and develop skills in identifying and communicating relevant population health information.

9 Class Participation:

Due to the comprehensive nature of the topics covered, attendance is mandatory for all class sessions. Students are expected to:

- Complete assigned readings before each session to facilitate meaningful discussions.
- Actively participate by sharing insights, opinions, and experiences to foster a collaborative learning environment.
- Contribute to discussions using tools such as *Poll Everywhere* for engagement and attendance tracking.

If absence is unavoidable, students should email me before the session.

10 Quizzes (35pts, $\approx 12\%$):

Seven short quizzes will be administered across the semester, reflecting on the topics we have covered, generally the week prior. These quizzes are not cumulative: each focuses on the material from the previous quiz forward. Each is worth **5 points**, and **all seven count — none is dropped**. Missing one costs you those points, so treat them as fixed appointments.

The quizzes are deliberately low-stakes and frequent: they exist to give you — and me — a running signal of what is landing and what is not. **Five fall before the midterm** (September 8 and 15, October 1, 15 and 20) and act as its running rehearsal. **Two fall after it — November 10**, on study design, and **November 19**, on planning and evaluating interventions. Those two exist for a simple reason: the material taught in November is as examinable as the material taught in September, and without a checkpoint it is the first thing forgotten.

Quiz Format & Timing:

- Quizzes are administered at the **beginning of class** on the dates listed in Table 2 and in the course schedule.
- Quiz duration: **5–7.5 minutes** (extended time for students with accommodations).
- Format: 3–5 questions, mostly multiple choice or short open-ended.
- **A one-page cheat sheet (one side, handwritten or printed) is allowed for every quiz.** Note that this is *not* the case for the midterm exam.

- Quizzes 6 and 7 fall on days when other work is also due; they are short, and the cheat sheet is allowed. Prepare the sheet, not an all-nighter.

Quiz Review:

- **Immediately after the quiz**, the instructor will review responses with the class.

Note that students who require a make-up quiz (or need another setting) will have the same format, but different questions.

11 Midterm Exam (50pts, $\approx 17\%$)

A **proctored, in-class midterm exam** will be held on **Tuesday, October 27, 2026**. It occupies the **entire class period** — the full 75 minutes. There is no lecture that day; come prepared to write for the whole session.

This is the single most heavily weighted assessment in the course. Unlike the quizzes, which sample recent material in small slices, the midterm asks you to hold the first half of the course together as one body of knowledge — and it is the one assessment taken under conditions I can fully control.

Format:

- **No notes, no cheat sheet, no electronic devices.** Blank scratch paper and a basic calculator will be provided.
- Multiple choice and true/false, plus several short open-ended and calculation questions (rates, standardization, life expectancy, OR/RR, sensitivity/specificity).
- Extended time and alternative settings for students with approved accommodations — please contact me at least two weeks in advance.

Coverage: everything from Week 1 through the causation material (Chapters 1–5): determinants of health, population health theories and frameworks, measuring health and disease, life tables and DALYs, health indicators and data sources, epidemiological measures and test characteristics, and causal inference.

Review: a review session will be held in class on **Thursday, October 22**, and a practice set will be posted on Course Site the week before.

Missed midterm. Because the midterm carries $\approx 17\%$ of your grade in a single sitting, it is the largest amount riding on one occasion in this course. If you cannot sit the exam on October 27, contact me **before** the exam whenever that is possible (travel, athletic competition, religious observance, court date — these are all foreseeable and all accommodated if raised in advance). In the case of illness or emergency, contact me as soon as you are able and provide documentation; a make-up exam will be scheduled within one week, in the same closed-book format but with different questions. **An unexcused absence from the midterm results in a score of zero**, so please do not simply not show up — talk to me.

12 Peer Presentation: Present a Classmate's Community (15pts, 5%)

Students will be assigned to groups of 2 (max 3). Each student will present **another student's community** (not their own) in a 2-3 minutes presentation. **Pairings are announced on September 8**, the first session after the add/drop deadline, leaving you four weeks to read your partner's material and prepare.

Your presentation should:

- Briefly introduce the peer's community (location, basic demographics)
- Identify key population health issues or concerns
- Highlight main determinants across levels (individual, community, macrosystem)
- Note particularly interesting findings or unique characteristics
- Acknowledge that even affluent communities face health challenges

This format mirrors real academic/professional settings where individuals often summarize others' work and provides valuable peer feedback before final paper submission. Presentations will take place on **October 6 and October 8, 2026**. The presentation (PPT, Google Slides, or other format) must be uploaded via Course Site, or the link emailed to me, by midnight on Monday, October 5 — whether you present on the Tuesday or the Thursday.

13 Individual Paper: Community Assessment (40pts, ≈13%)

You are expected to **conduct a critical assessment of your hometown community in terms of its multilevel health-promoting and health-inhibiting characteristics**. Your investigation may entail driving (or walking) around the community if you are currently in the area, reflecting on your previous experiences, interviewing others who live in the community, and internet research (e.g., county health rankings, publicly available community needs assessments, and other data) to learn more about the key characteristics of your county and/or state. If you are not sure whether there is sufficient data on your hometown to write about, you are welcome to look at a different scale (e.g. county, or local region) - but please talk to your instructor first.

You will write up your findings in an **8-page, double-spaced paper using 12-pt. Times Roman or Arial font with 1-inch page margins**.

Timeline:

- **Peer presentations: October 6 & 8, 2026.**
- **Final paper due November 24, 2026** (the last content session, before Thanksgiving).

14 Group Work: 65 pts

Milestones (required):

- Topics assigned by the instructor: **September 1, 2026**
- Project Scoping Note (0 pts, required): **September 30, 2026**
- Progress Checkpoint Report (7 pts): **November 12, 2026**
- Presentation slides due: **Monday, November 30, 2026 at midnight**
- Oral Presentation (32 pts): **December 1 & December 3, 2026**
- Final Written Report (26 pts): **December 7, 2026**
- Confidential peer evaluation (required): **December 7, 2026**

Topics

You do not choose your own topic. Each group is assigned a population health issue by the instructor when groups are formed on **September 8** — the first session after the add/drop deadline of September 4, since groups cannot be fixed until the roster is. Topics are presented in Week 1 so you know the range in advance.

This is deliberate. Left to choose freely, groups gravitate toward issues that are comfortable, over-documented, or too broad to analyse in twelve pages, and much of September disappears into deciding. An assigned topic puts every group on the same footing and sends you straight to the work that actually matters: finding the population, choosing the framework, and building the multilevel analysis. If your group has a serious objection to its assigned issue, come and see me within a week of the assignment.

What remains entirely yours: the **population of interest**, the **theory or model** you apply, the **evidence** you gather, and the **intervention** you propose. Two groups given the same issue will produce very different reports.

Project Scoping Note

The scoping note is a required formative submission. It confirms that your group has turned an assigned issue into a workable project — appropriately scoped, with a defined population and a suitable framework. Although ungraded, an approved scoping note is required to proceed. **Submission format:** 1 page (bullet points acceptable) **Required elements:**

- The assigned population health issue, restated in your own words
- Population of interest and why they are affected
- Selected population health theory/model (or 2 candidate options)
- One-sentence statement for each determinant level you plan to cover (individual, interpersonal, community/organizational, policy/macrosystem)

- Initial division of responsibilities (who will draft which determinant level/section)
- 2–3 credible starting sources (at least 1 peer-reviewed)

Feedback focus: scope, feasibility, clarity of population definition, fit of theory/model, and ability to address multilevel determinants within an 8–12 page report.

Progress Checkpoint Report

The checkpoint report is a structured progress review designed to ensure that groups are developing a coherent, evidence-based population health analysis that is on track for the final report. This is not a draft of the full paper, but a focused synthesis of work completed to date. **Submission format:** 3–4 pages (double-spaced), plus references **Required elements:**

- Clear restatement of the population health issue and population of interest
- Justification and brief explanation of the selected population health theory/model
- Preliminary synthesis of at least **two determinant levels** (e.g., individual and community), supported by peer-reviewed evidence
- Description of key data sources you are using or plan to use (e.g., surveillance data, census data, published studies)
- Brief outline of proposed intervention or policy implications aligned with the theory/model
- Updated division of group responsibilities and remaining tasks
- **One sentence per member stating what that member has personally done so far**

Expectations:

- Writing should be analytical and evidence-based, not descriptive
- Claims must be supported with citations (APA format)
- Figures or tables are optional but encouraged if they clarify key patterns

Feedback focus: conceptual coherence, appropriate use of theory/model, depth of determinant analysis, quality of sources, and feasibility of completing the final report at a high level.

Individual Contribution and Peer Evaluation

Group work is worth 65 points, and in previous semesters a minority of students have let their group carry them. That is unfair to the people who did the work, and this year it is addressed directly: **the group grade is a starting point, not automatically your grade.**

Three things make contribution visible.

- **Named sections.** Each determinant level is written and signed by one member, and the title page carries the responsibility matrix. Your section is your work and is read as such.

- **Live presentation.** Each member presents their own level and answers questions on it. Someone who has not engaged with the material is apparent within a minute.
- **Confidential peer evaluation.** With the final report, **each member submits a short confidential evaluation** of every member of the group, including themselves: what that person contributed, how reliably they met internal deadlines, and how the work divided in practice. This is a required submission.

How this affects your grade. Where the peer evaluations agree with each other and with what I observed — at the checkpoint, in the presentation, and in the report itself — I will adjust individual grades away from the group grade, **by up to 20% in either direction**. A member who carried the group can finish above it; a member who contributed little can finish well below it, and a member who contributed essentially nothing can receive close to zero for the group component regardless of what the group produced. A single dissenting evaluation will not move anyone's grade: adjustments require converging evidence.

If your group has a problem, raise it early. A group that tells me in October that a member has gone silent is a problem I can help solve. The same information delivered in the peer evaluation on December 7 is only a grievance — the report is already written, and the person who compensated has already done the extra work. The checkpoint report on November 12 asks each member to state what they have personally done so far; use it honestly.

15 Final Exam (30pts, 10%).

The final exam is **cumulative and covers the entire course**, from the determinants of health in Week 1 through spatial analysis in Week 14, in roughly even proportion. Think of it as a checklist rather than a gauntlet: it sweeps the whole semester at moderate depth, confirming that the pieces are still there and still connected, rather than probing any one topic hard. The midterm already did the hard probing on the first half of the course; quizzes 6 and 7 did it on the second.

The exam will be a timed online exam administered through CourseSite during the exam period: **December 8–16, 2026**. It consists primarily of **multiple choice and true/false questions**, with **a few open-ended questions** designed to assess deeper understanding.

A note on the weighting. At 30 points the final matters, but it is not where this course is decided — the in-class midterm, the seven quizzes, the individual paper and the group project together carry roughly three quarters of your grade. That balance is deliberate. As set out in Table 3, the final is unproctored: I cannot verify what you use while taking it, and I will not pretend otherwise. So the weight of the course sits on the assessments whose conditions are equal for everyone, and the final is asked to do something more modest — to confirm, at the end, that the semester holds together in your head.

What this means for November and December. Every topic taught after the midterm — study design, planning interventions, program evaluation, infectious disease modelling, and cartography and spatial analysis — is examinable on the final, and quizzes 6 and 7 will already have tested the first two of those in class. This is the material you are most likely to use in practice. Attendance and engagement in these weeks are expected exactly as they are in September.

Final Exam Review Session: A comprehensive review session will be held during the last week of classes.

16 Electronic Devices

Electronic devices – including laptop, cell phones, tablets etc. – should be in silent mode during class. This means silencing cell phones and suspending email, texting, and phone calling during class. Should a true emergency arise that you must address, please quietly attend to it and return to class as quickly as possible.

17 Writing Policies

You are expected to submit assignments using proper English grammar, syntax, and spelling. You are encouraged to use spell check and grammar check prior to submitting your written work. All written work must be **double-spaced in 12-pt, Times Roman or Arial font**. Please make sure you have the **margins set to 1 inch**. The Writing Center is available to anyone who may need assistance. Grammar, syntax, and spelling will account for 10% of the grade for written work. Also, you are expected to utilize the APA Publication Manual as a guide for writing papers for this course, and the grading rubric will be based on its precepts.

18 Use of AI Tools

Use AI as much as you want. I use it to prepare this course, and you will use it in whatever job you take next. There is no restriction on how much or how often you use language models or any other AI tool for the work you do outside of class in POPH 001 — as long as you say so.

What I care about is different: **you have to understand what you hand in**. That is not a moral position, it is a practical one. This course has two places where borrowed understanding becomes visible immediately. The **midterm exam** is closed-book, in class, with no notes and no devices, and **seven in-class quizzes** run from September to November — whatever you did not actually learn will show up there, and together they carry a quarter of your grade. And your **presentations are live**: in the peer presentation and the group presentation you will be asked questions about what you just said. Use these tools to learn faster, not to skip the learning.

There is also work here that AI simply cannot do for you. The individual paper asks you to assess **your own hometown** — the streets you know, the clinic your family uses, what closed down and what replaced it. A language model has never been there. It can help you structure the paper and find county-level data; it cannot supply the observation, and a paper built only from what it can generate reads exactly like one.

Per College of Health policy, AI use must be **disclosed**, AI-generated content must be **verified**, and you remain **accountable** for everything you submit. Disclosure is the one thing I require, and it is not a formality: every submission ends with a short **AI use note** stating **which** tool or platform you used, **what** you used it for, and **at which stage** of the work — finding sources, planning, analysing, interpreting, or writing. Two or three sentences is plenty. *No amount of disclosed use costs you points.*

Table 3 is the course-specific AI policy required by the College of Health.

Table 3: Instructor-specific AI policy for POPH 001

Course Task	AI Allowed	Conditions/Instructions	Enforcement
Assignments 1–5 (homework)	Yes	Unrestricted — explaining a concept, checking a calculation, drafting, editing. You must be able to explain every answer you submit.	The same material returns on the closed-book midterm, where borrowed answers do not help.
Health News presentation	Yes	Unrestricted for finding and summarising a story. The judgement of <i>why it matters for a population</i> must be yours.	You present live and may be asked what the story shows.
Peer presentation	Yes	Unrestricted for preparing slides. You are presenting a classmate’s community — read their work, do not have it summarised for you.	Presented live; expect questions.
Individual paper (community assessment)	Yes	Unrestricted for structuring, editing, and locating data. The observation of your own community must be genuinely yours.	Graded on the specificity of the community evidence, which is where generated text fails.
Group project (topic, checkpoint, report, presentation)	Yes	Unrestricted. All members must understand the whole report, not only their own determinant level.	Presented live in the final week; individual follow-up if only some members can explain the work.
Quizzes (in class)	No	No devices. A one-page cheat sheet is allowed — writing it is itself a study exercise.	In class, in view.
Midterm exam (in class)	No	Nothing at all: no notes, no cheat sheet, no devices. Scratch paper and a basic calculator are provided.	Proctored.
Final exam (online)	<i>Not possible</i>	The exam is unproctored, so I cannot verify what you use, and I will not pretend otherwise. This is one reason it is worth 30 points while the proctored midterm is worth 50. Sit it as an honest self-check: it is the last signal you get about what you actually know before you leave this course.	None possible — which is the point. The weight of your grade sits on the assessments where conditions are equal for everyone.
Disclosure & citation	Required	An “AI use” note in every submission: which tool or platform, what for , and at which stage . Quote and attribute any text taken verbatim.	A missing note just means you are asked to add it; repeated omission means a conversation.

19 Course Policies

19.1 Respect and Professionalism

Lehigh University endorses *The Principles of Our Equitable Community* ([link](#)). We expect each member of this class to acknowledge and practice these principles. Respect for each other and for differing viewpoints is a vital component of the learning environment inside and outside the classroom. You are expected to behave in a respectful and professional manner in this course. Disruptive and unprofessional behavior (rudeness, vulgar language, etc.) will result in warning, suspension, or dismissal from the course.

19.2 Academic Integrity

Students are bound by all sections of the *Lehigh University Student Code of Conduct*. You are responsible for making yourself aware and understanding the policies and procedures in the *Student Handbook* that relate to academic integrity. These policies include cheating, fabrication, falsification, forgery, multiple submission, plagiarism, conduct, complicity, and computer misuse. Please, see www.lehigh.edu/inprv/faculty/academicintegrity.html to access academic integrity resources.

19.3 Non-Discrimination

Lehigh University upholds *The Principles of Our Equitable Community* and is committed to providing an educational, working, co-curricular, social, and living environment for all students, staff, faculty, trustees, contract workers, and visitors that is free from harassment and discrimination. Such harassments or discriminations are not be tolerated.

The University strongly encourages (and, depending upon the circumstances, may require) students, faculty, staff, or visitors who experience or witness harassment or discrimination, or have information about harassment or discrimination in University programs or activities, to immediately report such conduct. If you experience, witness, or become aware of such behavior, you are encouraged to report the incident to:

- **Lehigh University Police Department (LUPD):** 610-758-4200
- **Equal Opportunity Compliance Coordinator/Title IX Coordinator (EOCC):** 610-758-3535 or eocc@lehigh.edu

Anonymous online reporting forms are available:

- *Gender Violence Incident Notification Form*
- *Discrimination, Harassment, Retaliation, or Bias Incident Reporting Form*

More details can be found at <https://eocc.lehigh.edu/>.

19.4 Accommodations for Students with Disabilities

Lehigh University is committed to maintaining an equitable and inclusive community and welcomes students with disabilities into all of the University's educational programs. In order to receive

consideration for reasonable accommodations, a student with a disability must contact Disability Support Services (DSS), provide documentation, and participate in an interactive review process. If the documentation supports a request for reasonable accommodations, DSS will provide students with a *Letter of Accommodations*. Students who are approved for accommodations at Lehigh should share this letter and discuss their accommodations and learning needs with me early in the semester as possible. For more information or to request services, please contact:

- **Disability Support Services:** Williams Hall, Suite 301
- Phone: 610-758-4152
- Email: indss@lehigh.edu
- Online: <https://studentaffairs.lehigh.edu/disabilities>

19.5 Religious Observances

If you have a religious observance that prevents you from doing work (e.g., quizzes, exams, or assignments) at its scheduled time, you will be given the opportunity to make up the work. You must notify the instructor at least **two weeks** prior to such an absence and propose how to make up for the missed academic work.

19.6 Notice Regarding Mandatory Reporting Obligations of Faculty

All faculty, staff, and administrators of Lehigh University are required by the **Policy on Harassment and Non-Discrimination** to report to the Equal Opportunity Compliance Coordinator/Title IX Coordinator and/or police any incident of bias, discrimination, harassment, retaliation, and sexual misconduct (including sexual harassment, sexual assault, stalking, dating violence, domestic violence, and sexual exploitation) that they become aware of, directly or indirectly, that involves students. Mandatory reporting helps to ensure that students are connected to the full range of resources and options afforded to them. If you would like to speak with someone confidentially (someone who does not have a reporting obligation to the University), you are invited to contact any or all of the following internal and external resources:

- University Counseling & Psychological Services, Johnson Hall, 610-758-3880
- University Chaplain's Office, Dialogue Center, 610-758-3877
- Crime Victims Council of the Lehigh Valley, 610-437-6611 (24-hour hotline)
- Turning Point of the Lehigh Valley, 610-437-3369 (24-hour hotline)

More information can be found at <https://eocc.lehigh.edu/> or by contacting the Equal Opportunity Compliance Coordinator & Title IX Coordinator, at eocc@lehigh.edu or at 610-758-3535.

20 Required Content for Individual Paper

1. Brief introduction

- (a) State the name of your local community. Please also specify which township/city, county, and state in which you reside.

2. Individual-level determinants of health

- (a) Describe your community in terms of its gender, age, race, and ethnic composition.
- (b) Discuss some protective (e.g., healthy diet, physical activity, positive screening behaviors, social distancing) and risky (e.g., smoking, unhealthy diet, substance misuse, etc.) health behaviors in which residents in your community (or county) typically engage.

3. Interpersonal-level and community-level determinants of health

- (a) Discuss health-promoting and health-inhibiting interpersonal-level determinants of health in your county. If you are unable to observe or find research on these determinants, this is understandable; please indicate if this is the case.
- (b) Discuss some of the social norms and policies in your neighborhood, workplace (or relative's workplace), and community in general.
- (c) Identify some community organizations that aim to address the broader determinants of health (e.g., food, transportation, housing, socioeconomic conditions, health care, etc.).
- (d) Identify and discuss some key stressors (e.g., crime, violence, poverty, racism/discrimination, etc.) in your study area.

4. Macrosystem-level determinants of health

- (a) Describe the socioeconomic profile (income level, level of education) of your community. How do you think this socioeconomic profile shapes the health of residents in your community/county?
- (b) Discuss the cultural profile (e.g., in terms of race and ethnicity, immigrant communities, etc.) of your county/community. How do you think this cultural profile may shape social norms in your county/community?
- (c) Describe some key aspects of the physical environment in your county/community and how they may shape health.
- (d) Discuss how politics may shape the health of residents in your community.
- (e) Discuss access to healthcare in your community, in terms of the distribution of insured/uninsured, use of public or private insurance, and physician-to-population ratio. Think about the other broader determinants of health. Describe one's access to health-promoting factors (e.g., affordable and healthy food sources, free or affordable opportunities for recreation, public transportation, affordable housing, and educational opportunities) in your community/county.
- (f) Think about the **DISTRIBUTION** of the health-promoting factors identified in 4e within your community/county. Do you feel that these resources are equitably distributed to most or all community residents? Why or why not?

5. Conclusion

- (a) Based on data (e.g., community health needs assessment, county health rankings), list the identified leading population health priorities in your community/county.
- (b) Discuss how this assignment has shaped your view of your local community/county.

21 Required Content for Group Presentation

Oral Presentation (15 minutes):

1. Brief Introduction

- (a) Topic of the presentation
- (b) Names of group members and category/level of determinant represented by each member

2. Background of Selected Health Issue/Disease

- (a) Describe the manifestation (i.e., signs and symptoms) of the selected health issue/disease.
- (b) Describe the prevalence of the health issue/disease in the US and/or globally.
- (c) Identify which part of the population the health issue/disease predominantly affects.
- (d) Discuss the short- and long-term impacts of the health issue/disease on the population.

3. Population Health Theory/Model

- (a) Describe the historical context (i.e., who developed it, when was it developed, etc.) of the selected population health theory/model.
- (b) Briefly describe the individual components/determinants that constitute the theory/model.
- (c) Briefly discuss the proposed relationships between the individual components/determinants and health/disease within the theory/model.
- (d) Briefly discuss the proposed interrelationships between the individual components/determinants within the theory/model.

4. Determinants of Selected Health Issue/Disease

- (a) Using the selected theory/model, discuss in detail how each category/level of determinants contributes to the selected health issue/disease. **EACH group member** is required to discuss the category/level of determinants he/she/they represent(s).
- (b) Imagine that your group is developing an intervention, policy, or research study to address or examine your selected health issue/disease. How would you **INTEGRATE** multiple determinants of health in your approach?

5. Citations

- (a) Please provide a list of citations at the end of the presentation.

22 Required Content for Group Written Report

The written report (8–12 pages, double-spaced) must be structured as follows:

1. **Title Page** (not counted in page length)

- Project title
- All group members' names
- Course information (POPH 001, Fall 2026)
- Date
- **Section responsibility matrix** – clearly indicate which member wrote which section

2. **Executive Summary** (1 page maximum)

- Concise overview of the health issue
- Key findings from your analysis
- Main recommendations

3. **Background and Significance** (1–2 pages)

- Detailed description of the health issue/disease
- Epidemiological data (prevalence, incidence, trends)
- Affected populations and health disparities
- Short- and long-term consequences
- Justification: Why is this issue important for population health?

4. **Theoretical Framework** (1–2 pages)

- Selected population health theory/model (e.g., Socio-Ecological Model, PRECEDE-PROCEED, Life Course Framework)
- Historical context and developers
- Key components and levels of the model
- Explanation of relationships between components
- Rationale: Why is this framework appropriate for your health issue?

5. **Multilevel Determinants Analysis** (3–5 pages total)

CRITICAL: Each group member must write 1–2 pages analyzing ONE level of determinants. Clearly label each subsection with the author's name.

- **Individual-Level Determinants** [Author: -----]
 - Biological and genetic factors
 - Health behaviors (diet, physical activity, substance use)
 - Psychological factors (stress, mental health)
 - How these factors contribute to the health issue
- **Interpersonal-Level Determinants** [Author: -----]
 - Social networks and social support
 - Family dynamics and relationships
 - Peer influence and social norms
 - Cultural factors and health beliefs

- **Community/Organizational-Level Determinants** [Author: -----]
 - Built environment and neighborhood characteristics
 - Access to healthcare and community resources
 - Schools, workplaces, and institutional factors
 - Community organizations and social capital
- **Policy/Macrosystem-Level Determinants** [Author: -----]
 - Health policies and regulations
 - Economic systems and socioeconomic factors
 - Political context and governance
 - Cultural values and societal norms
 - Environmental and structural racism
- **Integration Across Levels** [Collaborative section]
 - How do determinants at different levels interact?
 - Feedback loops and cascading effects
 - Upstream vs. downstream factors

6. **Intervention Approach** (1–2 pages)

- Description of proposed intervention
- How the intervention addresses multiple determinant levels
- Target population and setting
- Implementation strategy
- Expected outcomes
- Evaluation plan (process and outcome measures)
- Sustainability considerations

7. **Conclusion and Implications** (1 page)

- Summary of key insights from your analysis
- Implications for practice
- Implications for policy
- Limitations of your analysis
- Future research directions

8. **References** (not counted in page length)

- Minimum 8–10 peer-reviewed sources
- APA 7th edition format
- Use of primary sources preferred
- No Wikipedia citations

Formatting Requirements:

- 8–12 pages (excluding title page and references)

- Double-spaced
- 12-pt Times Roman or Arial font
- 1-inch margins on all sides
- Page numbers
- APA format for in-text citations and reference list
- Headings and subheadings to organize content
- Tables, figures, or graphics as appropriate (do not count toward page limit if placed in an appendix)

Individual Accountability: Each section analyzing a specific determinant level MUST be clearly attributed to its author. Use one of these formats:

- Section heading: "3.2 Individual-Level Determinants [Written by Jane Smith]"
- Footnote at beginning of section: "This section written by Jane Smith"
- Table on title page showing section assignments

This ensures that each group member's contribution is identifiable for grading purposes.

23 Pedagogical Spine and Course Philosophy

This course follows a carefully designed conceptual progression:

Structural Determinants → Measurement → Causation → Intervention → Modeling & Synthesis

This sequencing ensures that students understand **why** health outcomes are patterned (social and structural determinants) before spending extended time on **how** those patterns are measured and analyzed. By foregrounding determinants early, measurement becomes more interpretable, causal reasoning more grounded, and interventions more equity-focused. The Fall 2026 structure splits the semester into two halves with distinct centres of gravity:

- **First half (Weeks 1–10): individual mastery.** Five short, low-stakes quizzes with immediate in-class review build a running picture of what is landing, culminating in a proctored in-class midterm on October 27. Quizzes allow a one-page cheat sheet; the midterm does not.
- **Second half (Weeks 11–14): the applied material, still checked.** Students meet the parts of the field they are most likely to practise — designing studies, planning interventions, evaluating programs, modelling outbreaks, and mapping health outcomes — while the group project runs in parallel. **Two further quizzes** (November 10 and 19) keep this material live rather than leaving it to be revised once, in December, for an unproctored exam. The individual community assessment is due November 24, the last content session.
- **Final week (Week 15): synthesis in front of peers. All teaching is complete before the group presentations begin.** By December 1, students have the full toolkit — theory, measurement, causal reasoning, intervention design, evaluation, modelling and spatial analysis — and can draw on any of it in their presentations and in the questions they put to other groups. Presenting last is what turns the project from an assignment into a synthesis of the whole semester.
- **Assessment under equal conditions.** Roughly three quarters of the grade rests on work produced under conditions identical for every student — the proctored midterm, the seven in-class quizzes, the live presentations, and written work submitted under your own name. The unproctored online final is deliberately not the deciding assessment.
- **Interactive learning** – Continued emphasis on in-class exercises and application.
- **Current events connection** – Student health news presentations link course concepts to real-world developments.

A Note on Grading

Assignments and quizzes may be graded with the help of a course grader, working from a rubric I write and under my supervision. Everything else — the midterm, the individual paper, the presentations, and the group project — is graded by me. Whoever marked a given piece of work, **the responsibility for your grade is mine**: if something looks wrong or you want to understand a mark, come to me directly.

Where a quiz or assignment reveals a misconception shared by much of the class, I will address it with everyone in the following session rather than only in individual comments.

24 Tentative Course Schedule – Fall 2026

Week	Tuesday	Thursday
1	August 25, 2026 • Introduction to Public Health and Population Health – Introduction to the syllabus. Overview of course structure. – Introduction of each other. – Expectations RE: Peer presentations, group projects. – Presentation of the group project topics.	August 27, 2026 • Introduction to Public Health (chapter 1) – Prevention, promotion, and protection. – Historical evolution of public health. – Demographics: births and deaths • Read CNN article on life expectancy. • Read Chapter 1
2	September 1, 2026 • Structural Foundations & Determinants of Health – Social gradient and structural inequities – Social, economic, and environmental contexts • Groups are created and topics assigned. • Introduction to Assignment 1 (Social & Environmental Determinants).	September 3, 2026 • Health News (3 students) • Structural Foundations – Housing, policy, labor conditions. – Community health assessments vs. population health assessments • In-class group exercise: Identifying determinants in real communities • Read Chapter 4 (sections on determinants)
3	September 8, 2026 • QUIZ 1 (beginning of class) • Population Health Theories & Frameworks – Socio-ecological model – Life course perspective – Biopsychosocial model • Introduction to Assignment 2.	September 10, 2026 • Health News (3 students) • Theories & Frameworks – Pathways from determinants to health outcomes – How do various determinants interact? • In-class exercise: Identifying moderators and mediators on a causal pathway • Assignment 1 due Sep 11

Week	Tuesday	Thursday
4	September 15, 2026 • QUIZ 2 (beginning of class) • Health News (3 students) • Measuring Health and Disease in Populations – Rates, ratios, proportions – Incidence and prevalence * Population structure (age pyramids) * Demographic changes: fertility, mortality, migration * Preston curve: GDP vs. life expectancy • In-class exercise: age pyramids	September 17, 2026 • Health News (3 students) • Measuring Health and Disease in Populations – Demographic and health transition – Standardization in epidemiological research – Crude death rate – Standardized mortality rates • Read Chapter 2
5	September 22, 2026 • Research Skills Workshop – How to access health data and library resources – Explore datasets (CDC, WHO, County Health Rankings) – APA citation and writing format – Bibliography tools • Practice accessing datasets and analyze one indicator (e.g. HDI)	September 24, 2026 • Health News (3 students) • Measuring Health and Disease in Populations – Direct and indirect standardization: worked examples – Introduction to life tables • In-class exercise: standardizing mortality between two counties

Week	Tuesday	Thursday
6	September 29, 2026 • Health News (3 students) • Measuring Health and Disease in Populations – Constructing life tables and deriving life expectancy – Potential years of life lost – Combining mortality and morbidity into DALYs • Introduction to Assignment 3. • Read Chapter 3 • Assignment 2 due Sep 29 • Group scoping note due Sep 30	October 1, 2026 • QUIZ 3 (beginning of class) • Health News (3 students) • Health Indicators – Disease classification – Death certificates, health surveys, hospital records, disease registries – Reliability and validity of different data sources – Health indicators and indices
7	October 6, 2026 • PEER PRESENTATIONS – Day 1 • Slides due Mon Oct 5, midnight • Assignment 3 due Oct 7	October 8, 2026 • PEER PRESENTATIONS – Day 2
8	October 13, 2026 off-line • Group work – Meet in groups in person or on-line – Divide the determinant levels among group members – Identify sources of prevalence (e.g. by county, state) – Gather and screen peer-reviewed sources – Plan the work leading to the checkpoint report (Nov 12) • (Pacing Break is Monday Oct 12)	October 15, 2026 • QUIZ 4 (beginning of class) • Health News (3 students) • Epidemiological Measures – Sensitivity, specificity, and predictive values of diagnostic tests – Screening trade-offs • Exercise: Understanding sensitivity and specificity among covid patients • Introduction to Assignment 4

Week	Tuesday	Thursday
9	October 20, 2026 • QUIZ 5 (beginning of class) • Last quiz before the midterm • Health News (3 students) • Causation – Risk Assessment framework – Hazard identification, dose-response, exposure assessment – Measures of association: odds ratio, relative risk • In-class exercise: Calculating and interpreting OR/RR • Read Chapter 5 • Assignment 4 due Oct 21	October 22, 2026 • Causation (continued) – Bradford Hill criteria for causal inference – From association to causation • MIDTERM EXAM REVIEW SESSION – Practice questions and Q&A – Chapters 1–5
10	October 27, 2026 • MIDTERM EXAM • (the entire class period) – Full 75 minutes, 50 pts – No notes, no cheat sheet, no devices – Chapters 1–5	October 29, 2026 • Health News (3 students) • Study Design – Study designs (RCT, case-control, cross-sectional, cohort) – Ethical concerns (vaccines, randomization) – Validity, reliability, errors • Read Chapter 6 • Midterm returned and reviewed in class • Introduction to Assignment 5
11	November 3, 2026 • CIVIC ENGAGEMENT DAY – No Class	November 5, 2026 • Health News (3 students) • Study Design (continued) – Bias and confounding in observational studies • In-class exercise: Identifying potential confounders • Group project work time

Week	Tuesday	Thursday
12	November 10, 2026 • QUIZ 6 (beginning of class) • <i>covers study design (Ch. 6)</i> • Health News (3 students) • Planning Population Interventions – Types and levels of interventions – Criteria for screening programs – Behavioral models for health promotion – Cross-cultural issues • Read Chapter 7 • Assignment 5 due Nov 10 • <i>Last assignment of the semester</i>	November 12, 2026 • Health News (3 students) • Planning Interventions (continued) • In-class exercises: Planning interventions (tobacco cessation, greenways, food in schools) • Group Project Checkpoint report due Nov 12
13	November 17, 2026 • Health News (3 students) • Evaluating Health Programs – Framework for evaluations – Evaluation methods – Economic appraisal – Case studies • Read Chapter 8	November 19, 2026 • QUIZ 7 (beginning of class) • <i>covers interventions & evaluation (Ch. 7–8)</i> • Health News (3 students) • Emerging Tools: Infectious Disease Modeling – Compartmental models: SIR (and optionally SEIR) – Core logic, assumptions, and interpretation – What models can and cannot do for policy • Exercise: Simulating an outbreak • AI in Population Health – Prediction vs explanation – Targeting interventions, fairness and ethics

Week	Tuesday	Thursday
14	November 24, 2026 • Health News (<i>3 students</i>) • Cartography and Spatial Analysis in Population Health – Geospatial data & maps – Clusters, and ecological fallacy • <i>Last content session – all examinable material is now covered</i> • Individual Paper due Nov 24	November 26, 2026 • THANKSGIVING RECESS – No Class • (<i>Recess runs Nov 25–27</i>)
15	December 1, 2026 • GROUP PRESENTATIONS – Day 1 • Slides were due Mon Nov 30, midnight	December 3, 2026 • GROUP PRESENTATIONS – Day 2 • FINAL EXAM REVIEW + COURSE WRAP-UP – Practice questions and Q&A – <i>Cumulative: the whole semester, Weeks 1–14</i> – Course wrap-up and reflections • Group written report due Monday Dec 7 at midnight
	FINAL EXAM PERIOD: December 8 – December 16, 2026 Online via CourseSite – 30 pts (10%), cumulative, MC/TF + Short Answers	